# The band boundary:
the first nonabelian obstruction of the supply-chain inventory gerbe

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**Abstract**

A supply chain, abstracted to a small category $\mathcal{C}$, carries fungible inventory as local stock-data glued along logistics operations only up to a relabelling group $K = \mathrm{Aut}F$ of the stock fibre $F$; the gluing data is then a $\mathrm{Aut}F$-gerbe over the nerve $N(\mathcal{C})$. A companion note showed that a global inventory requires an abelian degree-two class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(\mathrm{Aut}F))$ to vanish, under the standing hypothesis that the comparison maps between charts are inner. We remove that hypothesis. Following Giraud, the obstruction is *staged*: beneath the abelian gerbe class lies the *band*, the outer-automorphism transport $\rho = q \circ \alpha$ of the relabelling group, a genuinely nonabelian datum whose class $\beta = [\rho] \in H^1(N(\mathcal{C}); \mathrm{Out}(\mathrm{Aut}F))$ must vanish before $[\mathfrak{g}]$ is even defined. We prove that $\beta = 0$ exactly when the transport is inner-valued up to an outer relabelling, so a strict global inventory exists iff *both* strata vanish. We then show the band is invisible to fungibility: for an $n$-element commodity fibre the band and the centre class are *both* forced to zero for every $n \geq 3$, $n \neq 6$, with the two low exceptions disjoint; the nonabelian phenomenon is a feature of gauge- or torsor-structured stock, not of interchangeable units. A single $(\mathbb{Z}/p)^2$-gauge loop realises a nonzero band with no abelian shadow, exhibiting the first genuinely nonabelian invariant of the inventory program.

## 1 Introduction

**The problem.** A supply chain is a small category $\mathcal{C}$: objects are nodes (warehouses, plants, ledgers), morphisms are logistics operations, composition is sequencing. Inventory is a presheaf-like datum glued along the morphisms. In the companion note [2] we treated the case where charts are compared by *equality* of counts and found a degree-one class $[\theta_R] \in H^1(N(\mathcal{C}); R)$; its sequel [3] refined the comparison to *isomorphism* of stock-states — lots are fungible, so two charts agree only up to a relabelling group $K = \mathrm{Aut}F$ of the inventory fibre $F$. The gluing data is then an $\mathrm{Aut}F$-*gerbe* over the nerve $N(\mathcal{C})$, and a global inventory exists only if the gerbe is neutral.

*Kodamai. Source, verification scripts, and this note live at `github.com/scotmacbeth/work-in-progress`. Prepared for the Kodamai AI-Mathematician grant (Applications and Theory pillars) and the Strathclyde mini-course *Tangent Categories and Containers* (13–16 October 2026). *Status:* the criterion (a) and vacuity (b) below are proved; the nonvacuous band (c) is an explicit, machine-checked witness. The note has had one referee pass, folded into the parent gerbe note [3], which is under re-review; it is *not* yet independently refereed in this standalone form.

**The gap.** The gerbe note [3] computed the neutrality obstruction as an *abelian* degree-two class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(\mathrm{Aut}F))$ — but only under a standing hypothesis it stated honestly and hedged: that the comparison automorphisms between charts are *inner*. For a nonabelian relabelling group this is a genuine restriction. What lives beneath it? When the transport between charts is allowed to apply an *outer* automorphism of $\mathrm{Aut}F$, the abelian class $[\mathfrak{g}]$ is not even well-defined: its coefficient group $Z(\mathrm{Aut}F)$ is twisted, and the whole computation presumes a prior datum has been trivialised. That prior datum is the subject of this note.

**The result.** The obstruction is *staged*, exactly as Giraud's classification of nonabelian gerbes predicts [5]. Beneath the abelian gerbe class sits the *band* (French *lien*): the outer transport $\rho := q \circ \alpha : \mathcal{C} \to B(\mathrm{Out}\mathrm{Aut}F)$, where $q : \mathrm{Aut}\, G \twoheadrightarrow \mathrm{Out}\, G$ is the outer-automorphism quotient and $\alpha$ is the chart-comparison transport. Its class $\beta := [\rho] \in H^1(N(\mathcal{C}); \mathrm{Out}(\mathrm{Aut}F))$ is a genuinely nonabelian degree-one datum, and it is the floor of the obstruction tower.

**Theorem 1** (Criterion; §3)**.** *The band is always a strict cocycle, so $\beta$ is well-defined* (Lemma 5)*. It vanishes, $\beta = 0$, if and only if the transport $\alpha$ is, up to an* $\mathrm{Out}(\mathrm{Aut}F)$*-relabelling of nodes, valued in inner automorphisms. Consequently* (Corollary 7) *a strict global inventory exists iff both strata vanish: the band $\beta = 0$* and *the abelian gerbe class $[\mathfrak{g}] = 0$, the latter being defined only once the former holds.*

So the hedged hypothesis of [3] is now a precise, checkable condition, and the gerbe theorem is its *band-trivial stratum*. Two further results map the phenomenon. First, it is invisible to plain fungibility:

**Theorem 2** (Vacuity; §4)**.** *For a bare $n$-element commodity fibre,* $\mathrm{Aut}F = S_n$*, the band* and *the centre class are both automatically zero for every $n \geq 3$ with $n \neq 6$. The two low-order exceptions are disjoint: $n = 6$ carries a possible band (*$\mathrm{Out}S_6 = \mathbb{Z}/2$*) but no centre class, and $n = 2$ carries a possible centre class but no band. No single commodity fibre exhibits both.*

Second, it is nonvacuous in general, and separates cleanly from the abelian story:

**Theorem 3** (Witness; §5)**.** *Over a single reversible loop $\mathcal{C} = B(\mathbb{Z})$ with gauge-structured stock* $\mathrm{Aut}F = (\mathbb{Z}/p)^2$*, any nonidentity transport has $\beta \neq 0$, while the abelian class is forced to zero because $N(\mathcal{C}) \simeq S^1$ is one-dimensional. The band is then the sole and nonzero obstruction.*

**The method.** Everything turns on the inner/outer short exact sequence of the relabelling group and the pointed nonabelian $H^1$ of a nerve. The criterion is an elementary Schreier argument; the vacuity is the classical arithmetic of $\mathrm{Aut}S_n$ and $Z(S_n)$; the witness is a one-line computation on $S^1$. No gerbe-theoretic machinery is invoked — the point of the note is precisely that the first nonabelian stratum of the inventory gerbe can be isolated and decided by hand, and every claim is backed by a finite machine check.

**Consequences.** The band is a new, prior, genuinely nonabelian invariant of inventory, complementary to the centre class: the band needs $\mathrm{Out}(\mathrm{Aut}F) \neq 1$, the centre class needs $Z(\mathrm{Aut}F) \neq 1$. Theorem 2 turns the applications hedge into a dichotomy — *commodity inventory carries no nonabelian obstruction; the phenomenon requires gauge/torsor-structured assets* — a quotable grant line. And it sharpens the parent program's placement of inventory as a distinguished $H^2$ home on $\mathrm{DCont} \simeq \mathsf{Cat}$ [1] by exhibiting the degree-one nonabelian datum that sits underneath.

**Outline.** Section 2 fixes the two structure groups, the band, and proves it is a cocycle. Section 3 proves the criterion and its corollary. Section 4 proves the vacuity and exhibits the $S_6$ curiosity. Section 5 gives the nonvacuous witness. Section 6 reports, at trust grade *computed*, the first base on which the two strata *couple*, and states the one open problem. Section 7 summarises what this does for the program.

## 2 The transport, the band, and the two structure groups

Let $\mathcal{C}$ be a small category (a directed container: objects are nodes, morphisms logistics operations), $N(\mathcal{C})$ its nerve, $F$ the inventory fibre, and

$$G := \mathrm{Aut}F$$

the relabelling group of the stock. Local inventory trivialisations are the local objects of a $G$-gerbe over $N(\mathcal{C})$; comparing two charts across a morphism $f : x \to y$ transports the structure group by an *automorphism* of $G$. Collecting these comparisons gives the *transport*

$$\alpha : \mathrm{Mor}\,\mathcal{C} \longrightarrow \mathrm{Aut}(G),$$

functorial *up to inner automorphism*: re-trivialising a chart conjugates the transport, so $\alpha_{gf}$ and $\alpha_g\alpha_f$ agree only modulo $\mathrm{Inn}(G)$. (The 2-cell coherence data that measures this slack is the abelian gerbe datum of [3], valued in $Z(G)$; here we need only that the slack is inner.)

Recall the *inner/outer sequence* of $G$:

$$1 \longrightarrow Z(G) \longrightarrow G \xrightarrow{\text{conj}} \mathrm{Aut}(G) \xrightarrow{q} \mathrm{Out}(G) \longrightarrow 1, \qquad \mathrm{Inn}(G) = \mathrm{im}(\text{conj}) = \ker q \cong G/Z(G). \tag{1}$$

**Definition 4** (Band)**.** The *band* (lien) of the gerbe is the composite outer transport

$$\rho := q \circ \alpha : \mathrm{Mor}\,\mathcal{C} \longrightarrow \mathrm{Out}(G).$$

The band is the sheafified, outer-automorphism-valued shadow of the transport. This is the folklore reading of Giraud's *lien* [5]: a band is a structure group well-defined up to isomorphism, where the isomorphism is itself well-defined only up to inner automorphism — so what survives globally is exactly the Out-valued data. It rhymes with the "visible up to inner automorphism" slogan of the monodromy tower in [2]. The point of passing to Out is that it launders away the slack:

**Nerve cohomology convention.** For a group $Q$ (constant coefficients), a *cocycle* in $H^1(N(\mathcal{C}); Q)$ is a functor $\mathcal{C} \to BQ$, i.e. a map $\gamma : \mathrm{Mor}\,\mathcal{C} \to Q$ with $\gamma(gf) = \gamma(g)\gamma(f)$ and $\gamma(\mathrm{id}) = e$; two cocycles are *cohomologous* iff related by a node-relabelling $\lambda : \mathrm{Ob}\,\mathcal{C} \to Q$ via $\gamma'(f\colon x \to y) = \lambda(y)\,\gamma(f)\,\lambda(x)^{-1}$. Then $H^1(N(\mathcal{C}); Q) = Z^1/\!\sim$, pointed by $\gamma \equiv e$; "$[\gamma] = 0$" means $\gamma \sim e$. For $Q$ abelian this is ordinary simplicial $H^1$; for $Q$ nonabelian it is a pointed set.

**Lemma 5** (The band is always a strict cocycle)**.** *$\rho$ is a genuine functor $\mathcal{C} \to B(\mathrm{Out}G)$: $\rho(gf) = \rho(g)\rho(f)$ and $\rho(\mathrm{id}) = e$. Hence $\beta := [\rho] \in H^1(N(\mathcal{C}); \mathrm{Out}G)$ is well-defined, regardless of the up-to-inner slack in $\alpha$.*

*Proof.* By construction $\alpha_{gf} = \iota \cdot \alpha_g\alpha_f$ for some $\iota \in \mathrm{Inn}(G) = \ker q$. Applying $q$ kills $\iota$: $\rho(gf) = q(\alpha_{gf}) = q(\iota)q(\alpha_g)q(\alpha_f) = \rho(g)\rho(f)$. Normalisation $\rho(\mathrm{id}) = e$ is immediate. The inner slack dies in Out, so $\rho$ depends only on $\alpha$ mod Inn. □

"Band trivial" — the standing hypothesis of [3] — means precisely $\beta = 0$.

## 3 The band-triviality criterion

**Theorem 6** (Criterion)**.** *$\beta = 0 \in H^1(N(\mathcal{C}); \mathrm{Out}G)$ if and only if the transport $\alpha$ is, up to an $\mathrm{Aut}\,G$-valued node relabelling, inner-valued: that is, iff there is a cocycle $\tilde{\alpha} : \mathcal{C} \to B(\mathrm{Aut}G)$, cohomologous to $\alpha$, with $\tilde{\alpha}(f) \in \mathrm{Inn}(G)$ for every morphism $f$. In particular, if $\alpha$ already lands in $\mathrm{Inn}(G)$ then $\beta = 0$; the theorem says $\beta = 0$ is exactly the gauge-invariant closure of that literal "trivial band".*

*Proof.* ($\Leftarrow$) Suppose $\tilde{\alpha} = \mu \cdot \alpha$ with $\mu : \mathrm{Ob}\,\mathcal{C} \to \mathrm{Aut}\,G$ and $\tilde{\alpha}(f) \in \mathrm{Inn}G = \ker q$ for all $f$. Put $\lambda := q \circ \mu : \mathrm{Ob}\,\mathcal{C} \to \mathrm{Out}\,G$. For $f : x \to y$,

$$\rho(f) = q(\alpha(f)) = q\big(\mu(y)^{-1}\tilde{\alpha}(f)\mu(x)\big) = \lambda(y)^{-1}\, q(\tilde{\alpha}(f))\, \lambda(x) = \lambda(y)^{-1}\lambda(x),$$

since $q(\tilde{\alpha}(f)) = e$. Then $\lambda(y)\,\rho(f)\,\lambda(x)^{-1} = e$, so $\rho \sim e$, i.e. $\beta = 0$.

($\Rightarrow$) Suppose $\beta = 0$: there is $\lambda : \mathrm{Ob}\,\mathcal{C} \to \mathrm{Out}\,G$ with $\rho(f) = \lambda(y)^{-1}\lambda(x)$ for every $f : x \to y$. Since $q$ is surjective by (1), choose lifts $\mu(x) \in \mathrm{Aut}\,G$ with $q(\mu(x)) = \lambda(x)$ at each node, and set $\tilde{\alpha}(f) := \mu(y)\,\alpha(f)\,\mu(x)^{-1}$. Then

$$q(\tilde{\alpha}(f)) = \lambda(y)\,\rho(f)\,\lambda(x)^{-1} = \lambda(y)\big(\lambda(y)^{-1}\lambda(x)\big)\lambda(x)^{-1} = e,$$

so $\tilde{\alpha}(f) \in \ker q = \mathrm{Inn}(G)$ for every $f$. And $\tilde{\alpha}$ is a cocycle, cohomologous to $\alpha$ via $\mu$: for $f : x \to y$, $g : y \to z$, $\tilde{\alpha}(gf) = \mu(z)\alpha(g)\alpha(f)\mu(x)^{-1} = \mu(z)\alpha(g)\mu(y)^{-1} \cdot \mu(y)\alpha(f)\mu(x)^{-1} = \tilde{\alpha}(g)\tilde{\alpha}(f)$, with $\tilde{\alpha}(\mathrm{id}) = e$. □

This is Giraud's stratum [5] read on the simplicial site $N(\mathcal{C})$: the band is the Out-reduction of the structure cocycle, vanishing exactly when the transport is inner up to coboundary. The argument uses only (1) and the nerve's $H^1$ — no bitorsor machinery (contrast [6]).

**A worked check on $D_4$.** Take $G = D_4$, the symmetry group of the four-lot ring. Then $\mathrm{Aut}(D_4) = D_4$ (order 8), $\mathrm{Inn}(D_4) = (\mathbb{Z}/2)^2$, and $\mathrm{Out}(D_4) = \mathbb{Z}/2$ — *both* the inner and the outer part are nontrivial, so the criterion has real content. Over the single reversible operation $\mathcal{C} = B(\mathbb{Z})$ (so Mor $= \mathbb{Z}$, $\alpha(t) = \varphi \in \mathrm{Aut}D_4$), a transport is relabel-reducible into Inn iff $\varphi \in$ Inn, and $[\rho] = [q(\varphi)] = 0$ iff $\varphi \in$ Inn; the two sides coincide for all eight automorphisms $\varphi$ (script `clause_a_D4_check.py`).

**Corollary 7** (The staged obstruction)**.** *A strict global inventory — a global relabelling making every transport a strict identity — exists if and only if both strata vanish:*

$$\textbf{\textit{(i)}}\quad \beta = 0 \in H^1(N(\mathcal{C}); \mathrm{Out}G) \quad \textit{and} \quad \textbf{\textit{(ii)}}\quad [\mathfrak{g}] = 0 \in H^2(N(\mathcal{C}); Z(G)),$$

*where the abelian class $[\mathfrak{g}]$ of [3] is defined only once* ***(i)*** *holds. The obstruction is genuinely staged: $[\mathfrak{g}]$ is a secondary invariant, defined in the kernel of the primary band. Thus the main theorem of [3] is exactly the band-trivial stratum of the full obstruction, and Theorem 6 supplies the prior stratum it had hypothesised away.*

*Proof.* A strict global inventory is a relabelling $\mu : \mathrm{Ob}\,\mathcal{C} \to \mathrm{Aut}\,G$ with $\mu \cdot \alpha \equiv e$. Projecting by $q$ trivialises $\rho$, so **(i)** is necessary. Given **(i)**, Theorem 6 lets us take $\alpha$ inner-valued, $\alpha = \mathrm{conj} \circ \bar{\rho}$ for a functor $\bar{\rho} : \mathcal{C} \to B(\mathrm{Inn}G)$; the gerbe is then the gerbe of lifts of $\bar{\rho}$ through $G \to \mathrm{Inn}G$ in (1), which by [3] is neutral iff $[\mathfrak{g}] = 0$. So **(i)**∧**(ii)** is necessary and sufficient. □

## 4 Fungible stock is doubly vacuous

Let $F$ be a bare $n$-element set of indistinguishable lots, so $G = \mathrm{Aut}(F) = S_n$.

**Theorem 8** (Vacuity)**.** *For every $n \geq 3$ with $n \neq 6$, generic fungible inventory carries neither a band obstruction nor a centre class:*

- ***band:*** $\mathrm{Out}(S_n) = 1$, *so* $H^1(N(\mathcal{C}); \mathrm{Out}S_n) = \{*\}$ *and* $\beta = 0$ — *for every* $\mathcal{C}$ *and every transport;*
- ***centre:*** $Z(S_n) = 1$, *so* $H^2(N(\mathcal{C}); Z(S_n)) = 0$ *and* $[\mathfrak{g}] = 0$.

*There are exactly two low-order exceptions, and they are* disjoint*:*

- $n = 6$ *is the lone band exception:* $\mathrm{Out}(S_6) = \mathbb{Z}/2$, *so a transport using the exotic outer automorphism can have* $\beta \neq 0$ — *but* $Z(S_6) = 1$ *still kills the centre class;*
- $n = 2$ *is the lone centre exception:* $S_2 = \mathbb{Z}/2$, $Z = \mathbb{Z}/2 \neq 1$, *so* $[\mathfrak{g}]$ *can be nonzero (the minimal* $B(\mathbb{Z}/2)$ *witness of [3]) — but* $\mathrm{Out}(S_2) = 1$ *keeps the band trivial.*

*No single fungible fibre exhibits both obstructions.*

*Proof.* $S_n$ is a *complete* group — centreless with every automorphism inner, hence Out = 1 — for all $n$ except $n = 2$ ($S_2 = \mathbb{Z}/2$ is abelian, Aut = 1, Out = 1 anyway) and $n = 6$, where $|\mathrm{Out}(S_6)| = 2$ (Rotman [7, Thms. 7.5–7.7]). And $Z(S_n) = 1$ for $n \geq 3$, $= \mathbb{Z}/2$ for $n = 2$. Feeding Out = 1 into Theorem 6 forces $\beta = 0$; feeding $Z = 1$ into the centre coefficient of [3] forces $[\mathfrak{g}] = 0$. The exceptional values $n \in \{2, 6\}$ are distinct and each triggers exactly one stratum, as tabulated. □

**The $S_6$ exception, explicitly.** $S_6$ is the unique symmetric group with an outer automorphism. It sends a transposition (cycle type $2\,1^4$) to a product of three disjoint transpositions (cycle type $2^3$) — a map that cannot be a conjugation, since conjugation preserves cycle type. It arises from the exotic transitive embedding $S_5 \hookrightarrow S_6$ (the action of $S_5$ on its six Sylow-5-subgroups). We verified an explicit witness (`out_s6.py`): $S_5$ acting on its six Sylow-5-subgroups gives a transitive embedding $\psi : S_5 \hookrightarrow S_6$ with image $H$ ($|H| = 120$, transitive); the action of $S_6$ on the six cosets $S_6/H$ is an automorphism $\Phi$ with $\Phi(0\,1\,2\,3\,4\,5) = [1, 4, 2, 5, 0, 3]$ and $\Phi(0\,1) = (0\,1)(2\,5)(3\,4)$, machine-checked to be a homomorphism and bijection sending every transposition to cycle type $2^3$ — hence outer. Exhaustive counts give $|\mathrm{Aut}(S_5)| = 120$ (Out = 1) and $|\mathrm{Aut}(S_6)| = 1440 = 2 \cdot 720$ (Out = $\mathbb{Z}/2$). So for $n = 6$ a transport applying $\Phi$ along some operation realises $\beta \neq 0$ while $[\mathfrak{g}] = 0$: a band obstruction with no abelian shadow. Honest caveat: this is a curiosity, not an applications driver.

**The applications boundary.** The nonabelian inventory phenomenon is invisible to fungibility. Three or more interchangeable units create no degree-two relabelling obstruction ($S_n$ centreless) and no band obstruction ($S_n$ complete), bar the isolated $S_6$ curiosity. Both nontrivial strata require *gauge/torsor-structured* stock: a fibre whose automorphism group has nontrivial centre (for the centre class) or nontrivial outer group (for the band). This is the economically meaningful regime — allocation defined up to a gauge group $K$ acting by translation ($\mathrm{Aut}F = K$, centre $Z(K)$, outer group measuring how the process twists the gauge) — not plain interchangeability.

## 5 A nonvacuous band

**Theorem 9** (Witness)**.** *There is a supply-chain datum $(\mathcal{C}, G, \alpha)$ with band $\beta \neq 0$. Take $G = (\mathbb{Z}/p)^2$ (an abelian gauge group of stock), $\mathcal{C} = B(\mathbb{Z})$ (a single reversible operation, the "weld loop"), and $\alpha(t) := A$ for a chosen $A \in \mathrm{Aut}(G)$ with $A \neq I$. Then $\beta = [\rho] = [A] \neq 0 \in H^1(N(\mathcal{C}); \mathrm{Out}G)$. Moreover the strata separate cleanly: the band is the sole and nonzero obstruction, while the centre class is forced to zero.*

*Proof. The groups.* For abelian $G$, conjugation is trivial, so $\mathrm{Inn}(G) = 1$ and $\mathrm{Out}(G) = \mathrm{Aut}(G)$. For $G = (\mathbb{Z}/p)^2$, $\mathrm{Aut}(G) = \mathrm{GL}_2(\mathbb{F}_p)$. Hence $q = \mathrm{id}$ and the band sees the *entire* transport: $\rho = \alpha$. (This is the opposite extreme from the inner-valued regime of [3]: here every nonidentity transport is non-inner.)

*The base.* $N(B(\mathbb{Z})) \simeq S^1$. A cocycle $\rho : B(\mathbb{Z}) \to B(\mathrm{Out}G)$ is determined by the single value $\rho(t) = A$ (the generator $t$ maps freely). A node-relabelling by $g \in \mathrm{Out}G$ sends $A \mapsto gAg^{-1}$, so

$$H^1(N(B(\mathbb{Z})); \mathrm{Out}G) = \{\text{conjugacy classes of } \mathrm{Out}G\},$$

and $[\rho] = 0$ iff $A$ is conjugate to $e$, iff $A = I$.

*The witness.* Choose any $A \neq I$; then $\beta = [A] \neq 0$. Concretely for $p = 2$, $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ (order 6, three classes $\{I\}$, three involutions, two order-3 elements); take the transvection $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ of order 2, so $[A] \neq [I]$ (script `band_boundary_verify.py`). Thus $\beta \neq 0$.

*Clean separation.* On $\mathcal{C} = B(\mathbb{Z})$ the nerve $N(\mathcal{C}) \simeq S^1$ is one-dimensional, so $H^2(N(\mathcal{C}); Z(G)) = 0$ for *any* coefficients: the centre class is always zero here, no matter how rich $Z(G) = G = (\mathbb{Z}/p)^2$ is. Hence the band $\beta \neq 0$ is the unique obstruction, and it is genuinely nonabelian-flavoured — it lives in $H^1$ with nonabelian coefficients $\mathrm{GL}_2(\mathbb{F}_p)$, prior to and independent of any $H^2$ story. □

**Interpretation.** Stock is defined up to a $(\mathbb{Z}/p)^2$-gauge (two independent mod-$p$ ledger offsets). A cyclic refurbishment process $t$ relabels the gauge by a fixed $\mathrm{GL}_2(\mathbb{F}_p)$ element $A$. If $A \neq I$, no global choice of gauge is consistent around the loop — and this failure is a band obstruction, an outer twist of the structure group, strictly prior to the degree-two "how-many-units" gerbe class. Fungibility can never produce it (§4); gauge-structured stock does, minimally.

## 6 Where the strata couple

The two strata are *staged* (Corollary 7) — the band must vanish before the centre class is defined — but staging is not the same as independence. On a one-dimensional base they decouple completely: $H^2(S^1; -) = 0$, so in the witness of §5 every band is realised by a strict transport and the degree-two datum is vacuous. The first base on which the strata can genuinely *interact* is the torus $T^2 \simeq N(B(\mathbb{Z}^2))$, where $H^2(T^2; -) \neq 0$.

Here the relevant connecting map is that of the coefficient extension (1). For a band $\rho : \mathbb{Z}^2 \to \mathrm{Out}G$ with $G$ of nilpotency class $\leq 2$ (so $M := \mathrm{Inn}G = G/Z(G)$ is abelian), the pointed exact sequence of (1) on $T^2$ gives a connecting map

$$\delta : \; H^1(T^2; \mathrm{Out}G) \longrightarrow H^2(T^2; \mathrm{Inn}G),$$

and $\delta(\beta)$ is the obstruction to lifting the outer band to a strict $\mathrm{Aut}G$-valued transport. A short computation (the single relator of $\mathbb{Z}^2$ evaluated on the fundamental 2-cell) identifies $\delta(\beta)$ with the commutator $[A, B]$ of any lifts $A, B \in \mathrm{Aut}G$ of $\rho(s), \rho(t)$, taken in the coinvariants $M/((\sigma - 1)M + (\tau - 1)M)$; it is zero iff some lift choice makes $A, B$ commute. The problem of coupling therefore

reduces to a finite search: find a class-2 group with two commuting outer automorphisms whose lifts never commute.

**Remark 10** (Computed witness; not yet refereed)**.** Such a group exists: $G = D_4 \times C_2$ (order 16). With the two outer automorphisms $A : r \mapsto r^3,\ s \mapsto s,\ c \mapsto r^2c$ and $B : r \mapsto r^3c,\ s \mapsto s,\ c \mapsto c$ (both of order 2 in $\mathrm{Out}G \cong$ a group of order 16, generating a $(\mathbb{Z}/2)^2$), the band $\rho(s) = [A],\ \rho(t) = [B]$ has $\beta \neq 0$, and $\delta(\beta) = [A, B] = [s] \neq 0$ in $H^2(T^2; \mathrm{Inn}G) = (\mathbb{Z}/2)^2$: an exhaustive check over all $16 \times 16$ inner adjustments finds no commuting lift (scripts `aut_engine.py`, `verify_final.py`, independently re-verified). This is the first base on which a nonzero band *forces* a nonzero degree-two class — the two strata do not merely stack, they talk. *Three honest caveats.* (1) Trust grade is *computed*, not refereed, and not yet Lean-checked. (2) The coefficient of $\delta$ is the inner group $\mathrm{Inn}G$, *not* the centre $Z(\mathrm{Aut}F)$ (for $D_4 \times C_2$ the two happen to be abstractly isomorphic, but they are different functors of $G$ and sit in different extensions); the centre-valued phenomenon is a separate stage, the central extension $1 \to Z(G) \to G \to \mathrm{Inn}G \to 1$, which the same group also witnesses. (3) Minimality over all groups of order $\leq 16$ is not claimed.

**The open problem.** Classifying $K$-*banded* inventory gerbes — the torsor over the $\beta$-twisted $H^2(N(\mathcal{C}); Z(G))$, Giraud's full nonabelian machinery [5, 6] — is open and is the live research lane. Remark 10 settles existence of coupling with one explicit witness; the structured classification, and a Lean re-run of the finite certificate on the existing reduced-nerve $\mathbb{F}_2$ engine [4], remain to be done.

## 7 What this does for the program

- The standing "band-trivial" hypothesis of [3] is now a precise, checkable condition: $\beta = [q \circ \alpha] = 0 \in H^1(N(\mathcal{C}); \mathrm{OutAut}F)$ (Theorem 6).
- The main theorem of [3] — strict global inventory $\iff [\mathfrak{g}] = 0$ — is the band-trivial stratum of a staged obstruction; the complete statement is $\beta = 0$ *and* $[\mathfrak{g}] = 0$ (Corollary 7).
- For fungible stock the hypothesis is automatic for all $n \geq 3$, $n \neq 6$, and the centre class is also vacuous. The applications hedge becomes a theorem: *commodity inventory carries no nonabelian obstruction; nontriviality requires gauge- or torsor-structured assets* (Theorem 8).
- The band is a new, prior, genuinely nonabelian invariant, nonvacuous in general (Theorem 9) and complementary to the centre class: the band needs $\mathrm{Out} \neq 1$, the centre class needs $Z \neq 1$. It is the first place in the supply-chain program where genuinely nonabelian cohomological data appears.

**Bottom line.** The inventory gerbe's obstruction is staged, matching Giraud exactly over $N(\mathcal{C})$. First the band $\beta = [q \circ \alpha] \in H^1(N(\mathcal{C}); \mathrm{OutAut}F)$ must vanish — which happens iff the transport is inner-valued up to an outer relabelling (Theorem 6). Then the abelian class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(\mathrm{Aut}F))$ must vanish (Corollary 7). Fungible stock sees neither stratum for all $n \geq 3$, $n \neq 6$, with $n = 6$ (band only) and $n = 2$ (centre only) the disjoint exceptions (Theorem 8). And the band is nonvacuous: a single $(\mathbb{Z}/p)^2$-gauge loop with transport $A \neq I$ has $\beta = [A] \neq 0$ while its centre class is forced to zero (Theorem 9) — the first genuinely nonabelian content of the supply-chain program, prior to and cleanly separated from the abelian $H^2$ story.

## References


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